SpeechCARE Explainability Framework: Enhancing Transformer Interpretability for Cognitive Impairment Detection –NIA Alzheimer's Challenge

SpeechCARE Team

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1. Abstract. SpeechCARE is a novel, accessible, non-invasive, and cost-effective system for the early detection of cognitive impairment in three diagnostic categories: Control, Mild Cognitive Impairment (MCI), and Alzheimer's disease (AD). The model architecture utilizes pre-trained multilingual acoustic and linguistic transformers alongside a novel fusion mechanism that dynamically assigns modality-specific weights (acoustic, linguistic, and demographic) to improve the prediction outcome. This approach provides insight into the relative importance of each modality and thereby enhances interpretability. On the PREPARE challenge, SpeechCARE achieved a promising AUC-ROC= 0.88 and F1-score (a harmonic mean of precision and recall) =0.71 for distinguishing among Control, MCI, and AD. To address the inherent "black box" limitation of Transformer models, SpeechCARE employs an explainability framework that integrates advanced visualization techniques (e.g., spectrogram analysis,[1] SHAP[2]) and leverages state-of-the-art large language models[3] (LLMs) for deeper interpretability.

By designing an Explainer Panel, we facilitated understanding of SpeechCARE's decision-making process, making it accessible to individuals (e.g., patient, caregiver) with limited technical or clinical expertise. This Panel comprises three main components: A pie chart illustrating the contribution of each modality to the model decision making; and two components for highlighting and interpreting linguistic and acoustic cues. These components elucidate how specific speech cues correlate with cognitive impairment, raising awareness of one's cognitive status and supporting timely intervention. To demonstrate this capability, we selected three English-speaking participants (one from each diagnostic class) who completed the picture description task; SpeechCARE correctly identified the three participants according to their diagnosis. English was chosen for its broad audience reach in the United States. Note that due to the multilingual nature of the underlying Transformer models and the flexible fusion architecture, the explainability framework can be readily adapted to other languages and speech tasks.

2. Technical approach.

2.1. Summary of the SpeechCARE pipeline and performance. SpeechCARE consists of: (a) Preprocessing phase including audio quality evaluation and noise reduction, LLM-assisted speech task identification, and language detection/transcription; (b) Model architecture built on multilingual pre-trained transformers (mHuBERT,[4] mGTE [5]) and a novel Adaptive Gating Fusion (AGF) that dynamically weights acoustic, linguistic, and demographic features, enhancing performance across different speech tasks (e.g., picture description, see Figure 1). (c) Performance measurement, where SpeechCARE achieved promising AUC-ROC = 0.88 and F1-score = 0.72 for the three diagnostic classes (Control, MCI, AD). Although transformer embeddings provide rich representations, they are often treated as "black boxes." The SpeechCARE Explainability Framework addresses this by integrating visualization methods (SHAP[2], spectrogram analysis) and an LLM-based interpretation (LLaMA [6] 3.1 70B) to highlight informative linguistic/acoustic cues and interpreting them in a human-readable manner.

2.2 Explainability framework components.

Component A: Visualization of modality contributions. The SpeechCARE fusion framework assigns specific weights to each modality (acoustic, linguistic, and demographic), reflecting their contributions to the model's decision-making process. These weights are visualized in a pie chart within the Explainer Panel, offering insights into the relative importance of acoustic and linguistic features. Additionally, for a broader perspective at the dataset level, **Figure 1** provides insight into the average contribution of each modality to the model's decision

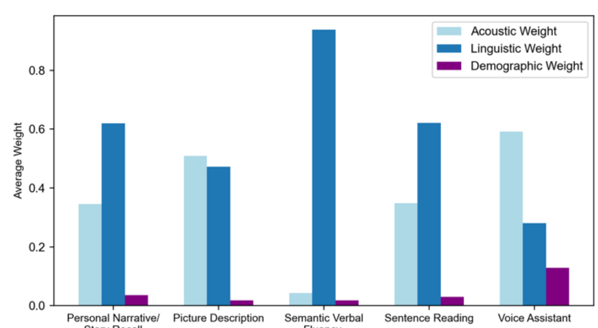

Figure 1. Average contribution of each modality to the model's decision making for each speech task

making for each speech task. The type of the speech task was identified using SpeechCARE LLM-assisted speech task identification module (see section 2.1).

Component B. Linguistic transformers explainability—visualizing linguistic cues linked to cognitive impairment: This component has two complementary explanation layers. See sections (a) and (b). Output of this component is transcriptions of audio files with highlighted informative linguistic cues, and an LLM-based, human-understandable interpretation of the cues **(See Explainer Panel).**

(a) Visualization of linguistic cues using SHPA[2] (Shapley Additive exPlanations): We integrated SHAP with the linguistic transformer model (mGTE [5]) to analyze how each token affects the model prediction. SHAP treats the model as a black box, systematically masking tokens and observing changes in the prediction. This process generates SHAP values, quantifying each token's contribution. By examining token-level importance scores, the model can identify subtle linguistic indicators (e.g., filler words) that may signal cognitive decline. SHAP also groups tokens that collectively shift predictions, highlighting broader linguistic patterns. Overall, output of SHAP includes: (1) A list of SHAP values assigned to tokens, (2) text with highlighted linguistic cues.

(b) Utilizing an LLM to provide higher level of explainability: For local, HIPAA-compliant deployment, we fine-tuned LLaMA[6] 3.1 70B using QLoRA[7] (a parameter-efficient, low-rank adaptation method) on the dataset. Hyperparameter tuning was conducted on the validation dataset (20% of the training data, see model report). No test dataset was used for fine-tuning. Using techniques of prompt engineering,[8] we developed a structural prompt with four components to instruct the fine-tuned LLaMA (**Table 1**) to identify linguistic cues—indicative of either healthy cognition or cognitive impairment—based on token-level SHAP values. The prompt intentionally did not include the predicted outcome or true label, enabling the model to infer relevant linguistic cues without introducing bias.

Table 1. Prompt components for SHAP-driven linguistic cue analysis using LLaMA (**Component B)**
Component 1. LLaMA as a cognitive impairment assistant: LLaMA serves as a specialized language model, trained to detect linguistic cues of cognitive impairment.
Component 2. Token-level attribution and contextual interpretation: LLaMA receives tokens' SHAP value, with higher values indicating a greater effect on the prediction outcome. It then analyzes each token's position in the text, linking it to other tokens or sentence structures.
Component 3. Linguistic indicators mapping: LLaMA receives a list of linguistic indicators including lexical richness (e.g., token range), syntactic complexity (simple vs. complex), repetition (e.g., consecutive-word repetition), disfluencies, fluency markers (e.g., "um," "uh,"), and semantic coherence (e.g., logical consistency). LLaMA is asked to map each influential token to the relevant indicator(s).
Component 4. Generating human-readable summaries: LLaMA links token-level cues to broader linguistic deficits, providing human-readable information for clinical interpretation.

Component C. Speech (acoustic) transformers explainability—visualizing acoustic cues linked to cognitive impairment: This component has two complementary explanation layers: **(a)** Audio spectrogram[1] that visualizes the energy of frequency components over time; **(b)** SHAP[2] -based spectrogram that highlights time spans that are informative for the predicted outcome (**Table 2**); **(c)** Using explainable acoustic and temporal features, we provide deeper insights into how important acoustic/temporal features relate to cognitive status **(Table 3)**. The output of this explainability component is a spectrogram visualization that highlights informative acoustic and temporal features, accompanied by human-interpretable explanations of these features **(See Explainer Panel).**

3. Discussion of limitations and uncertainty.

Despite offering rich insights into how acoustic, linguistic, and demographic features contribute to predictions, the explainability framework has several important limitations and sources of uncertainty. First, methods like SHAP provide short windows (a token or sequence of a limited number of tokens) embedding-level explanations but may oversimplify the complex, non-linear interactions within large transformer models, potentially overlooking subtle cues. Second, aggregating analyses (e.g., 300 ms segments in spectrograms) can mask fine-grained temporal features critical in early-stage cognitive impairment detection. Third, the reliance on demographic data and potential sampling biases could inadvertently lead to reduced generalizability across different populations. Fourth, while the framework has been validated on the PREPARE dataset, performance may vary when applied to real-world clinical settings with differing linguistic, cultural, or audio conditions. Finally, fine-tuning large LLMs for local, HIPAA-compliant deployment does not guarantee perfect interpretability, as model outputs are influenced by the scope and representativeness of training data, necessitating continuous refinement and external validation.

Table 2. Procedures of SHAP-based spectrogram highlighting **(Component C)**
Step 1. Embedding generation: We used mHuBERT's convolutional encoder to generate hidden representations (embeddings) for each audio file. Each embedding corresponds to a 25 ms interval.
Step 2. SHAP analysis. We applied SHAP to analyze how individual embeddings influence predictions. SHAP systematically masks embeddings, observes prediction changes, and computes Shapley values quantifying each embedding's contribution relative to a baseline (all embeddings masked).
Step 3. Spectrogram visualization: We used SHAP values to highlight voice segments indicative of cognitive impairment cues. First, we plotted the Mel-Spectrogram of the voice and mapped each 25 ms embedding to its corresponding time span on the spectrogram. Then, for each 25 ms, we computed: $\boldsymbol{opacity\ score}\ (25\ ms) = \ embedding\ SHAP\ value \times\ Spectrogram\ Energy$ Higher SHAP values increase opacity score, while lower values blur the spectrogram time span, highlighting each time span's importance to the predicted outcome. Since energy remains nearly constant in shorter intervals, SHAP values primarily determine opacity.
Step 4: Interpretability enhancement: We aggregated consecutive 25 ms time spans of the spectrogram into 300 ms blocks—approximating syllable production time in older adults—and computed the average opacity score for each block.

Table 3. Key acoustic features reflecting cognitive decline selected to provide a higher level of explainability (**Component C)**
Fundamental Frequency[9] **(F0, *visualized on the spectrogram*):** Provides insight into the control of the vocal fold; cognitive impairment increases F0 variability due to compromised motor planning.[10]
Formant Frequency[11] **(F3, *visualized on the spectrogram*):** Provides insight into the speaker's ability in articulatory coordination of tongue and lips; significant fluctuations in F3 may indicate compromised articulatory precision, reflecting potential cognitive decline.[12]
Pauses before nouns[13] (***visualized on the spectrogram****)*: Using an automated part-of-speech tagger, we identified nouns. Next, we extracted their timestamp and aligned them with the spectrogram to highlight preceding pauses, appearing as dark bands on the spectrogram. Cognitive impairment often leads to difficulty in lexical retrieval, particularly nouns.
Rhythmic structure[11] **(*visualized separately)*:** To provide insight into speaker rhythm, we used Shannon Entropy which measures variability in the voice signal; Cognitive impairment often leads to lower entropy, which indicates reduced voice signal complexity, perceived as a monotonous voice. [11]
Shimmer [14]: Provides insight into the short-term variability in the voice amplitude; higher Shimmer indicates an unstable voice. Cognitive impairment affects amplitude stability. We computed Shimmer's quartiles, resulting in four categories: very unstable, unstable, almost stable, and stable.[15]
Sound Pressure Level[16] **(SPL)**: Provides insight into vocal energy. SPL quantifies acoustic energy relative to the lowest hearing threshold (0 dB); in cognitively impaired speech, SPL often decreases or varies less, yielding a quieter or less energetic voice. We computed SPL quartiles, resulting in four categories: low energy, mild energy, moderate energy, and energetic.[17]

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